JalSakshi

Per-storm evidence that Delhi's rainwater-harvesting pits still work

Team: ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 1:** IoT-based underground water management

‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

A rule Delhi funds, checked by periodic inspection

What	Status	Source
RWH mandatory for plots ≥ 100 m²	in force **since 2001**	Indian Express (2026)
10 % water-bill rebate if functional (15 % with recycling)	in force	Hindustan Times, 21 Apr 2026
₹204.46 crore rebates paid, 2017–2024	reported	HT, citing DJB data
Functionality certificate	issued **after inspection**, periodic renewal	HT, 21 Apr 2026
Annual owner self-certification	**being planned**	HT, 21 Apr 2026
1.5× bill for non-compliance	**"may attract"**	HT, 21 Apr 2026

> "Till now, there was a rule but no real checking." — Delhi water minister (HT, 21 Apr 2026)

"Many have become non-functional due to inadequate maintenance" — an official, quoted by Indian Express

Why it matters now

- Delhi extracts **92.1 %** of its annually extractable groundwater. **10 of 34 tehsils** are over-exploited (337.44 km²).

  CGWB, *Dynamic Ground Water Resources of India 2025*, §7.32

- DJB Phase I planned to **restore 1,000 non-functional** RWH systems in government buildings and build 500 new ones (target was 30 Jun 2026; we found no completion report). 7,596 of 9,148 government buildings have RWH.

  Indian Express (2026)

- The campaign's recharge potential is **estimated at ≈ 733 million litres a year** (rooftop area × average rainfall). No instrument reports how much the structures actually take in.

  Indian Express (2026); an estimate, not a measurement

JalSakshi: one witness per recharge pit

IN THE PIT · PLANNED HARDWARE 0–5 m vented stainless steel probe (±0.5 % FS) in a slotted, socked stilling tube, **tip 5–10 cm above the floor** (water below the tip isn't seen)
IN THE LID · PLANNED MCU + LoRaWAN radio, signed local log; vent tube ends in a desiccant chamber; antenna clear of any metal cover. **Rain gauge required** (it wakes the node and checks plausibility)

BUILT · SOFTWARE, TESTED IN SIMULATION per storm:

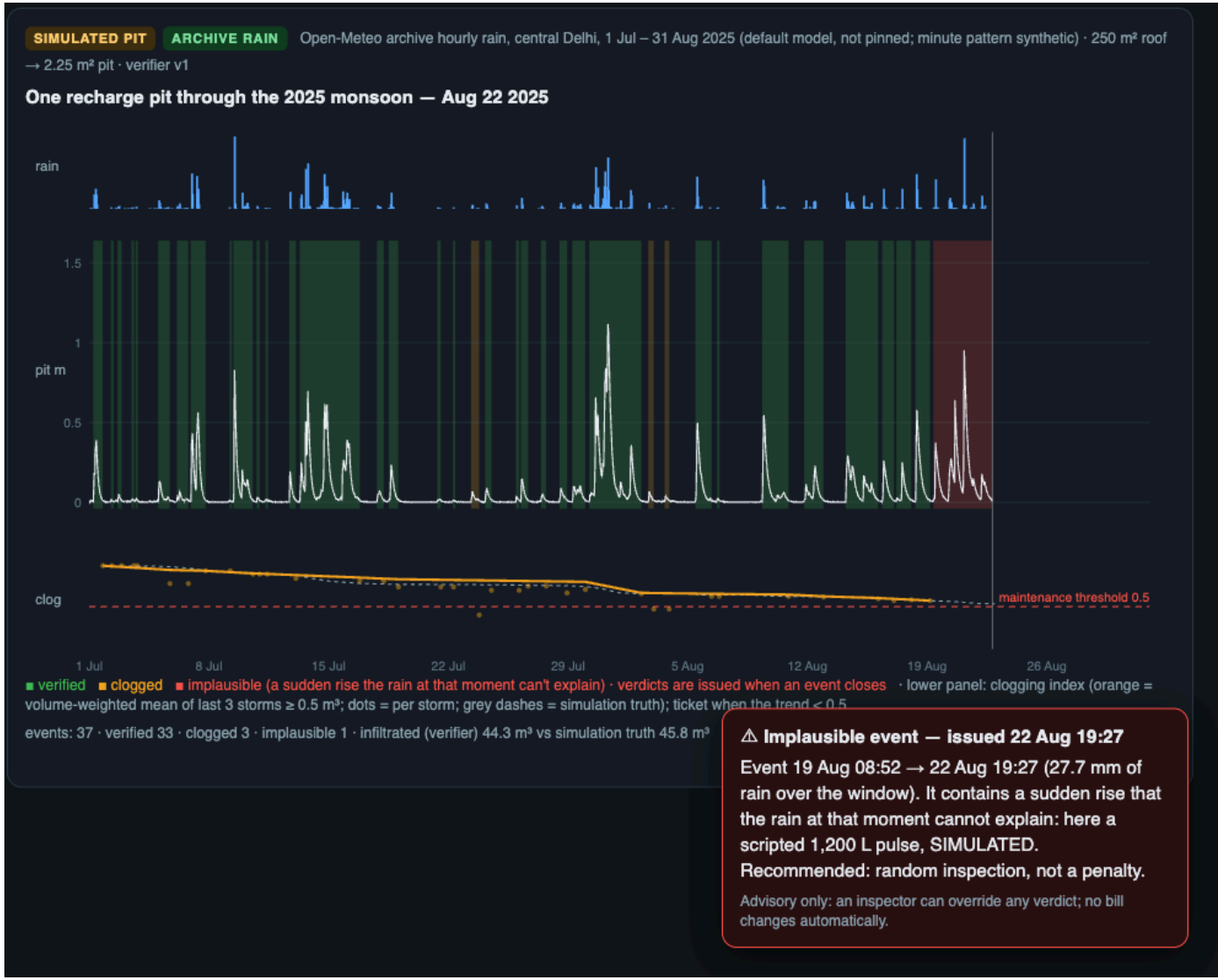
1. **Drain rate → clogging index** vs commissioning: the core product, **maintenance screening**
2. **Infiltrated volume: a model-based estimate** (storage + drain law fitted from one metered hose fill). Bypass and water leaving the pit are not observed. A per-event uncertainty band is **PLANNED**, not built.

Verdicts: verified · clogged · no inflow · implausible · **deferred** (missing rain or stage data → no judgement)
→ **ranks inspections; an inspector can override; no bill changes automatically**

60-second demo (offline laptop)

REAL Delhi DWLR network → **SIMULATED** pit on **archive** 2025 rain → a verdict when each event **closes** → tickets

- "Implausible" here = a **large injected pulse** (1,200 L in 1 min)
- A 20 L bucket (~9 mm rise) is below the 2 cm threshold: **not detected**
- Tickets arrive when an event closes: **hours to days** after onset
- 🔌 **Unplug rain gauge → deferred**, no accusation



How good is it? Simulation today; blind rig tests in November

40 simulated pits per split · monsoon-2024 holdout · rain: Open-Meteo archive (default model, not pinned); minute pattern synthetic

Frozen test	Median error	1 in 10 off by more than
Stress: non-linear drains + imperfect gauge	**13.1 %**	58 %
+ realistic probe: ±25 mm offset, drift, 2.1 mm steps	**14.5 %**	61 %
Same drain model as the verifier (best case)	3.0 %	16 %

- Unit: **event windows** with ≥ 0.2 m³ true infiltration (1,233 on the stress test)
- **Large injected pulses** (500–2,000 L): **10/10** flagged · honest windows flagged: 2/1,291 (stress), 15/1,156 (probe)
- Clogging precision / recall: **holdout 0.67 / 1.0** (10 pits) · **dev 0.40 / 0.67** (3 pits)
- Assumes **1-min sampling** throughout; the duty cycle is untested

Built for Delhi conditions: status, not promises

- PLANNED **Underground link:** only the probe is below ground; the antenna sits clear of the metal lid; a site survey picks LoRaWAN or NB-IoT
- DATASHEET **Accuracy:** 0.5 % of a 5 m range = ±25 mm per reading → ±0.11 m³ worst case on a 2.25 m² pit (A). The 5–10 cm below the probe tip (0.11–0.23 m³) is not seen
- PLANNED **Vent & silt:** vent end kept dry with desiccant; socked tube and stand. Auto-zero was tested in simulation, not adopted
- PLANNED TEST **Power & outages:** settle time and sleep current measured in Oct (D-cell vs solar); 24 h outage recovery is an acceptance test. **No battery-life claim**
- SAFETY **No pit entry:** probe lowered from the lid by an empanelled installer; low-voltage battery device; lids stay closed; install only with site permission
- PLANNED **Before each monsoon:** a metered-fill check, because clogging that develops in the dry season shows up only at the next storm

Cost, prospective customer, adaptation

Parts only (6 listings, 4 Oct 2026; excl. tax, shipping, mounting, labour, service)	₹
Base node (probe ₹4,254.51 + LoRa module + host board + cell + IP67 box + gland)	**11,166.18**
First site: node + rain gauge + shared gateway	48,454.18
3-pit pilot: 3 nodes + 3 gauges + 1 gateway	78,244.54

- For scale: one packaged CGWA-compliant DWLR (Acorn) lists ≈ ₹38,000 and logs twice a day. Not like-for-like.
- **Prospective pilot customer:** DJB's Phase-I restoration programme and its 59 empanelled RWH providers (IE). **No contact yet.** Then campuses and RWAs.
- **Adapting to other cities:** same software; per-site metered-fill calibration and safety check; local radio choice; local rebate rules. The first pilot covers **one structure type** (roof-runoff pit with filter media).

Impact, boundaries and the ask

- **Structure-level evidence** against the campaign's ≈ 733 M L/yr *estimate*: drainage and infiltration at structures, **not aquifer recharge**
- **Clogging alerts** after each observed storm, plus a pre-monsoon fill check, so crews clean the pits that stopped working
- **Water-quality gate:** pilot only on permitted roof-runoff structures with existing pretreatment and a responsible site operator. We never recommend recharging runoff we can't vouch for.
- **Roadmap:**
  - Oct: drum pour rig + node
  - Nov: blind weighed-water trials + 1–3 real pits
  - 1 Dec: report + working prototype
- **Ask:** a pilot partner (DJB zone, campus or RWA) and permission to instrument 3 pits

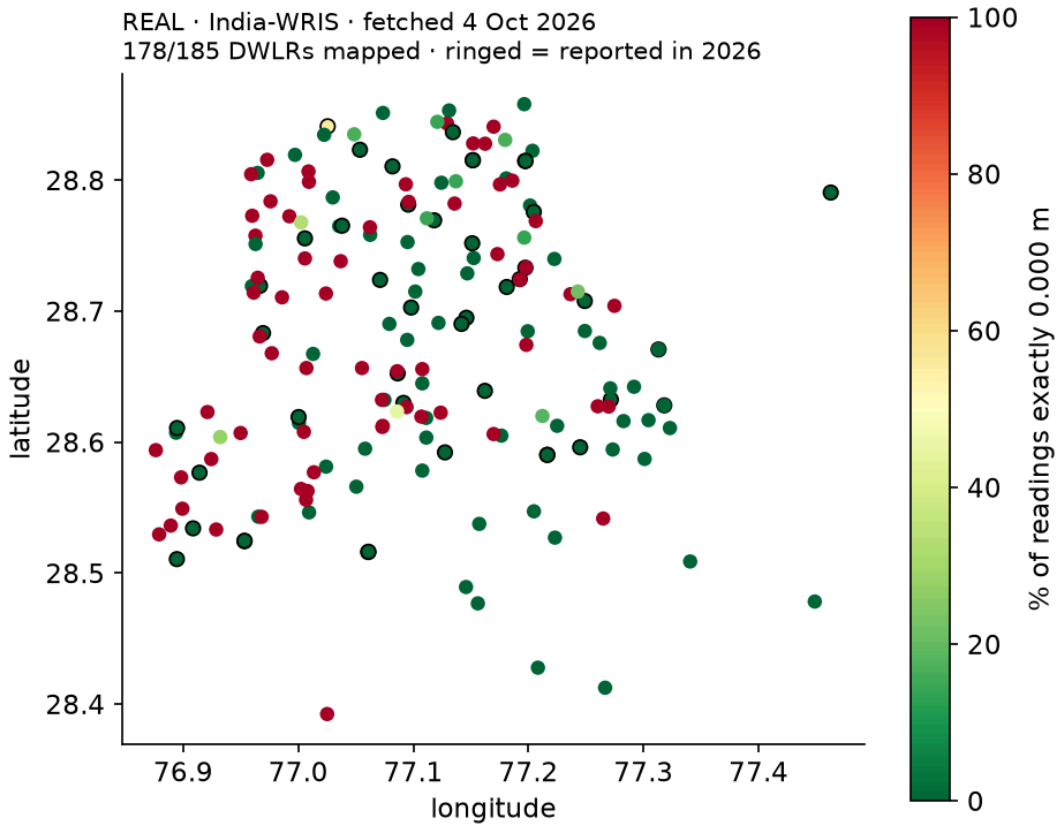
Measured, not self-declared.

Backup slides

Backup 0: Delhi's own sensors, a public-stream snapshot (REAL data)

India-WRIS public API, fetched 4 Oct 2026 (483,253 Delhi readings):

- **185** telemetric piezometers (DWLR), reporting every 6 h
- **46** reported in 2026 (latest public reading **4 Jun 2026**)
- **23 %** of readings are exactly 0.000 m; **17 %** of 6-h slots missing (up to each station's last report)
- Exploratory: median 2025 monsoon depth change **2.5 m** (79 stations); not attributed to RWH



Sources on slides or in notes · (A) = assumption · status labels: BUILT · SIMULATED · PLANNED

Backup 1: Why we don't claim to catch illegal borewells

- DJB identified **20,297** illegal borewells; **15,962** sealed; **4,000+** "couldn't be traced on the ground" (HT, 10 Jun 2025; Millennium Post). The NGT found seals that let borewells run "very conveniently" (The Bar Bulletin, 2026).
- Public piezometers report **every 6 h**, so pump cycles shorter than ~12 h alias. Drawdown is confounded by legal pumping, barometric pressure and rain.
- So we **never accuse a well**. A zone-level influence index is research only.

Backup 2: BOM and price evidence (4 Oct 2026)

Part	Vendor	₹
DFRobot KIT0139 0–5 m 4–20 mA probe	element14 India	4,254.51
RAK3172 LoRaWAN module (IN865)	electronicscomp.com	1,672.80
ESP32-C6-DevKitM-1 (prototype host)	robu.in	1,099.00
Forte ER34615 D Li-$SOCl_2$ 3.6 V	robu.in	879.00
IP67 ABS box 150×100×70 mm	amazon.in	3,250.87
PG7 gland	robu.in	10.00
DFRobot tipping-bucket rain gauge (listed unavailable)	amazon.in	3,729.00
shared: LPS8N 8-ch LoRaWAN gateway (IN865 variant unconfirmed)	amazon.in	33,559.00

Seen by us on 4 Oct 2026 (URLs: prototype-plan.md §3). An independent re-check the same day was blocked by most vendors (HTTP 403/503), so get quotes before buying. Not priced: boost converter, antenna, tube, sock, mounting, potting, cable.

Backup 3: The evaluators and the keep rule

- **v1:** linear-reservoir pits (same model as the verifier); **v2:** drain ∝ h^β, β ~ U(0.5, 1.5), gauge × lognormal(0.25)/day, 0–5 min lag; **v3:** v2 + probe offset ±25 mm, ±10 mm drift, floor at 0 m, 2.1 mm quantisation
- **Fleet:** 40 pits per split (healthy / clogging / blocked / faker); registry 5–10 % wrong. Dev = monsoon 2025, holdout = monsoon 2024.
- **Keep rule:** improve the target by > 0.5 points; worsen no other evaluator by > 10 %; confirm on holdout.
- **Chronology:** the v1 and v2 rules were committed before their results (git: 08:03 → 08:15, 08:26 → 09:46 IST). The v3 rule was written before its runs (our log) but committed together with them, so it isn't independently provable.
- **Pre-registered?** Read the holdout numbers as recorded checks, not an untouched prospective test.

Backup 4: One worked event, and what a level sensor can't see

One event (SIMULATED pit, archive rain)	Value
Window	30 Jul 05:53 → 2 Aug 07:01, 2025
Rain in window × 250 m² roof × 0.75 runoff prior	52.0 mm → 9.75 m³ (cross-check)
Infiltrated volume, model-based estimate	**9.80 m³**
Simulation truth	10.38 m³ (–5.6 %; best case, same drain model)
Fleet error, stress test	median 13.1 %, 1 in 10 off by > 58 % (**not** a per-event interval; that is PLANNED)

Water path	Seen by stage + rain?
Water entering the pit	**Yes**: the rise, checked against rain × roof
Inlet bypass / diverted downpipe	**Partly**: under-delivery vs rain × roof, or "no inflow"
Overflow	That it happened (time at spill level): **yes**; how much: **estimated**
Leak or crack out of the pit	**No**: indistinguishable from infiltration → inspection
Water below the probe tip	**No**: a 0.11–0.23 m³ blind zone (A: 2.25 m² pit)